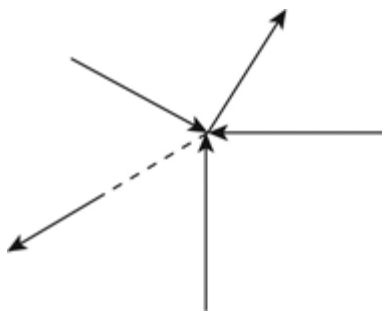
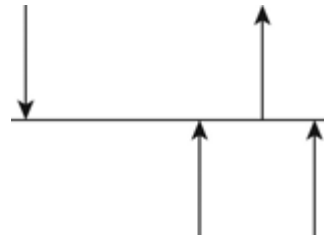


Topic 8 Summary: Statics

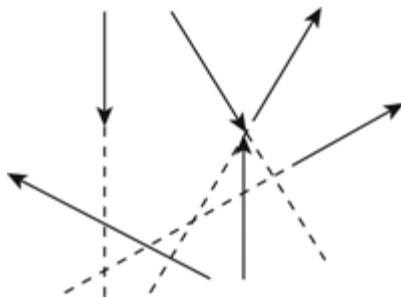
- What is statics?
- Definitions
 - Coplanar systems of forces
 - Concurrent forces
- The fundamental theorem of statics
- Resultant of any number of forces in one plane
- Bodies in equilibrium under a system of concurrent forces
- Theorem: the triangle of forces
- Lami's theorem
- The polygon of forces/ resultant of any number of forces
- Worked examples



Concurrent forces

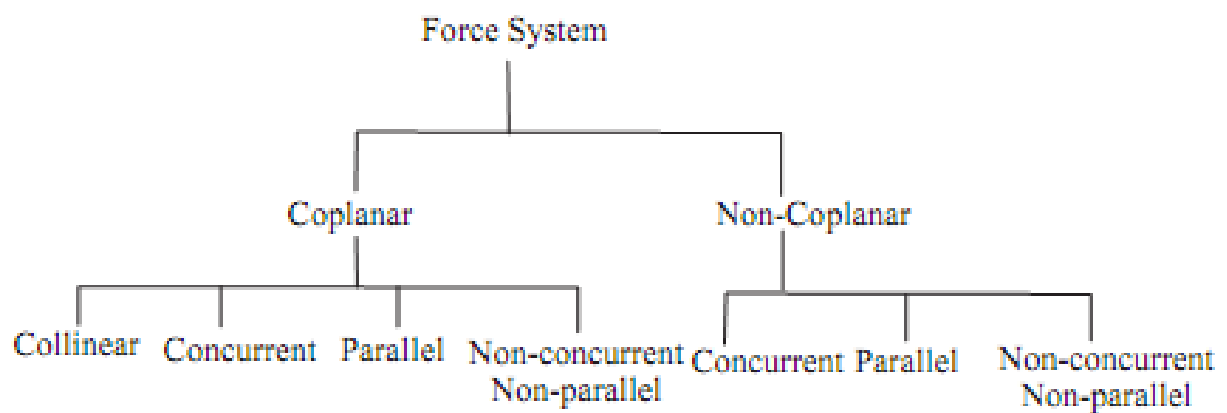


Non – concurrent forces
which are parallel



Non – concurrent
Non - parallel

NB: In all the above cases, the forces are coplanar.



STATICS

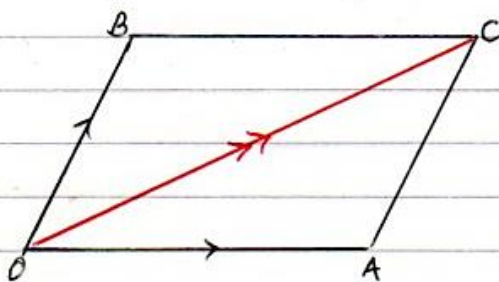
Statics is that branch of mechanics in which we study the effect of forces on bodies that remain at rest.

Definition: **Coplanar systems of forces** are systems in which all the forces can be represented as acting in the plane of the paper one is using.

If all these forces are such that all their lines of action pass through a single point, then the forces are **concurrent**. A concurrent system can always be reduced to a **single resultant force** which may be zero, but this is not the case for all non-current systems.

The fundamental theorem of statics (parallelogram of forces).

If two forces $\vec{P}$ and $\vec{Q}$, acting on a particle at O , can be represented in magnitude and direction by straight lines $\vec{OA}$ and $\vec{OB}$ (see fig.), then the effect of these two forces on the particle is equivalent to a single force which is represented in magnitude and direction by the diagonal $\vec{OC}$ of the parallelogram $OACB$.



$$\begin{aligned}\vec{OC} &= \vec{OA} + \vec{OB} \\ &= \vec{OA} + \vec{AC} = \vec{OB} + \vec{BC}\end{aligned}$$

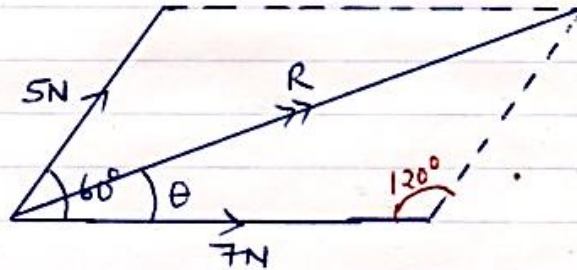
The single force $\vec{OC}$ is called **the resultant** of $\vec{OA}$ and $\vec{OB}$.

The resultant of two forces is **zero** only if the two forces are equal in magnitude and opposite in direction.

Example 1:

Two forces of 5N and 7N act at a point. Both are directed away from the point and the angle between them is 60° . Find the magnitude and direction of the resultant.

Sol:



By the cosine rule:

$$R^2 = 7^2 + 5^2 - 2(7)(5)\cos 120^\circ$$
$$= 109$$

$$\Rightarrow R = 10.44 \text{ N.}$$

To find θ , i.e. the direction of the resultant, we use the sine rule:

$$\frac{\sin \theta}{5} = \frac{\sin 120}{10.44} \Rightarrow \theta = 24.5^\circ$$

Thus R is 10.44N at angle 24.5° from the 7N force.

Example 2:

Forces of 3N and 5N, acting towards a certain point have a resultant of 6N. Find the angle between the forces.



$$\cos \alpha = \frac{3^2 + 5^2 - 6^2}{2 \times 3 \times 5}$$

$$\Rightarrow \alpha = 93.82^\circ$$

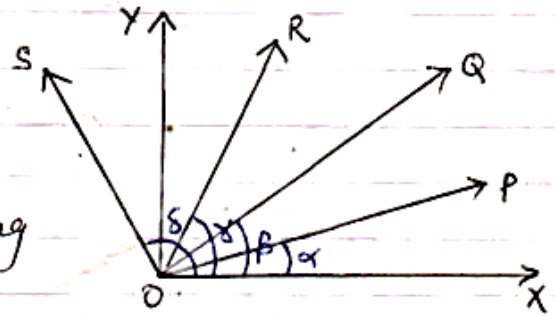
$$\therefore \theta = 180 - 93.82 = \underline{\underline{86.18^\circ}}$$

Resultant of any number of forces in one plane

Let the forces $P, Q, R, S, \dots$ etc act on a body at a point O . We draw through O two axes, OX and OY at right angles to each other. We assume that all forces act in the XY -plane.

Let $P, Q, R, S, \dots$ etc make angles $\alpha, \beta, \gamma, \delta, \dots$ with the line OX .

Resolving these forces along OX and OY , we get their components along these directions.



Hence, these forces are equivalent to a component

$$P \cos \alpha + Q \cos \beta + R \cos \gamma + S \cos \delta + \dots \quad \text{along } OX$$

and

$$P \sin \alpha + Q \sin \beta + R \sin \gamma + S \sin \delta + \dots \quad \text{along } OY.$$

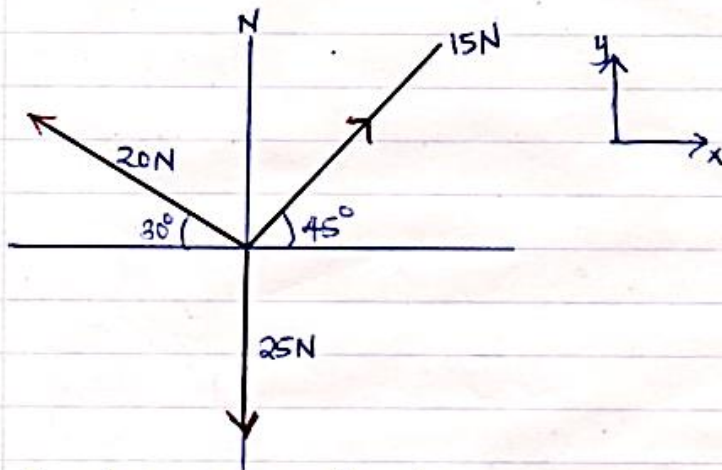
If these components are denoted by X and Y respectively and let $\vec{F}$ be their resultant which makes an angle θ with OX , then

$$F \cos \theta = X \quad \text{and} \quad F \sin \theta = Y$$

$$\therefore F = \sqrt{X^2 + Y^2} \quad \text{and} \quad \tan \theta = \frac{Y}{X}.$$

Example 2:

Coplanar forces of 15N, 25N, and 20N act away from a certain point on bearings of 045° , 180° and 300° . Find their resultant in magnitude and direction.



Resolving in x-direction

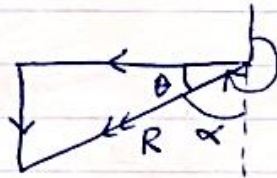
$$15 \cos 45 + 0 + (-20 \cos 30) = -6.714$$

Resolving in y-direction

$$15 \sin 45 + (-25) + 20 \sin 30 = -4.393$$

$$\therefore R = -6.714 \hat{i} - 4.393 \hat{j}$$

$$|R| = \sqrt{(6.714)^2 + (4.393)^2} = \underline{\underline{8.024 \text{ N}}}$$



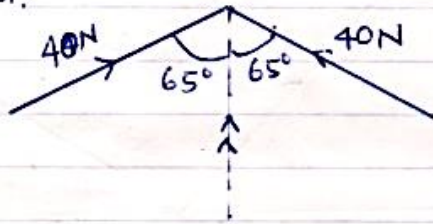
$$\tan \theta = \frac{4.393}{6.714} = 33.2^\circ \Rightarrow \alpha = 180 - 33.2 = 56.8$$

i.e. at a bearing of $180 + 56.80 = \underline{\underline{236.80^\circ}}$

Example 4:

Find the resultant of two forces of 40N each, acting towards a certain point, which are inclined to one another at 130° .

Sol:



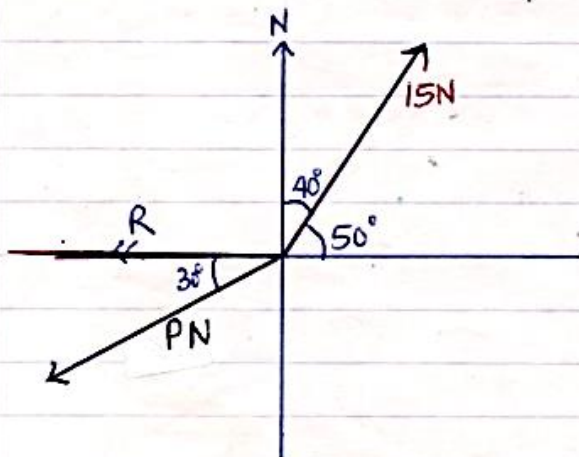
Resultant acts along line of symmetry. Components in this direction for each force is $40\cos 65^\circ$.

$$\therefore R = 2 \times 40\cos 65^\circ \\ = \underline{\underline{33.81 \text{ N}}}$$

Example 5:

Horizontal forces of 15 and P act away from their point of application on bearings of 040° and 240° . Given the resultant R is due West in direction, find P and R.

Sol:



Resolving into components

$$15\text{N} \rightarrow \begin{pmatrix} 15\cos 50^\circ \\ 15\sin 50^\circ \end{pmatrix}$$

$$P\text{N} \rightarrow \begin{pmatrix} -P\cos 30^\circ \\ -P\sin 30^\circ \end{pmatrix}$$

$$R\text{N} \rightarrow \begin{pmatrix} -R \\ 0 \end{pmatrix}$$

$\therefore$ Vertical components

$$15\sin 50^\circ - P\sin 30^\circ = 0 \Rightarrow P = \underline{\underline{22.98 \text{ N}}}$$

Horizontal component

$$\begin{aligned} R &= P \cos 30 - 15 \cos 50 \\ &= 22.98 \cos 30 - 15 \cos 50 \\ &= \underline{\underline{10.26 \text{ N}}} \end{aligned}$$

Bodies in Equilibrium under a System of Concurrent Forces

Forces in equilibrium are either parallel or concurrent (i.e. they either form a couple or pass through a single point).

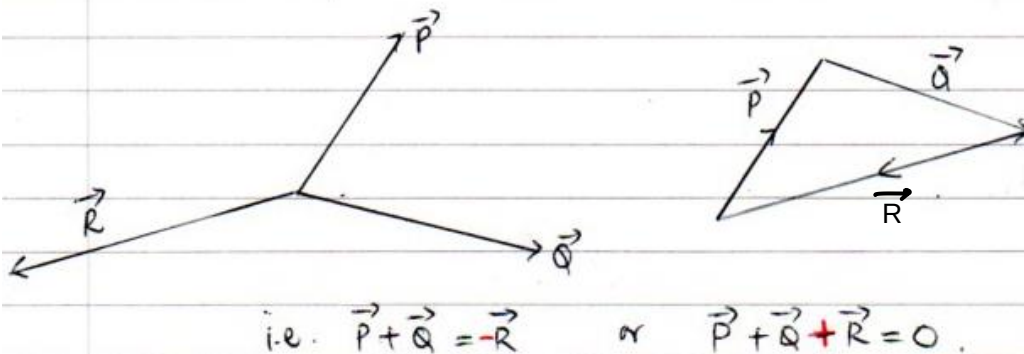
For concurrent forces to be in equilibrium, the resultant force must be zero. (i.e. if we resolve these forces in any 2 perpendicular directions, then the algebraic sum of their components must be zero).

Conversely, if the sums of their components in two perpendicular directions are zero, then the forces are in equilibrium.

The Triangle of Forces (Theorem)

If 3 forces acting on a body can be represented by the 3 sides of a triangle taken in order, then they will be in equilibrium.

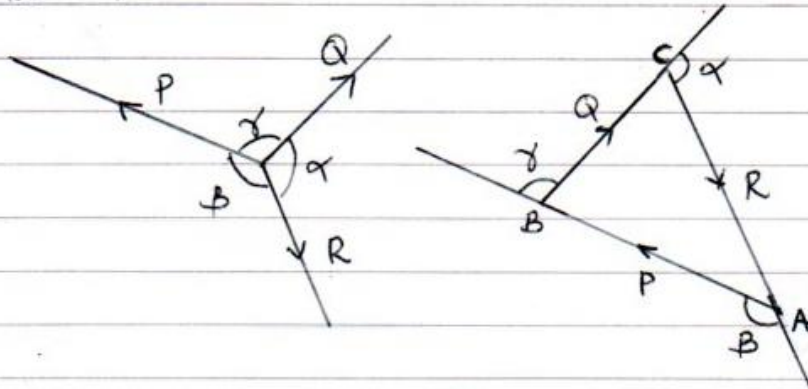
Consider 3 forces $\vec{P}$, $\vec{Q}$, and $\vec{R}$ that are in equilibrium



Conversely, if 3 forces $\vec{P}$, $\vec{Q}$, and $\vec{R}$ acting at a point are in equilibrium, they can be represented by the sides of a triangle whose sides are parallel to the directions of the forces.

Lami's Theorem.

If 3 forces acting at a point are in equilibrium, then each is proportional to the sine of the angle between the other two.



Let P , Q and R be the 3 forces that are in eqbm, and the corresponding triangle of forces be ABC . Let α , β , and γ be angles between Q and R , R and P , P and Q respectively. Applying the sine rule to the vector triangle ABC , we get

$$\frac{P}{\sin(180-\alpha)} = \frac{Q}{\sin(180-\beta)} = \frac{R}{\sin(180-\gamma)}$$

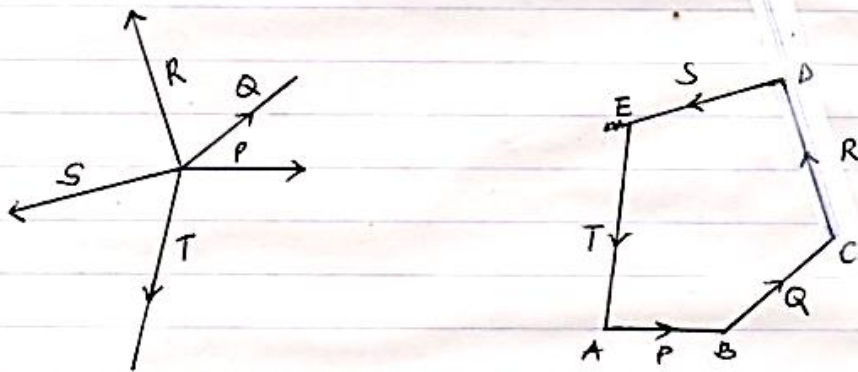
But $\sin(180-\theta) = \sin \theta$, $\therefore$

$$\boxed{\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}} \rightarrow \text{Lami's Thm.}$$

The Polygon of Forces.

If any number of forces, acting on a body can be represented in magnitude and direction by the sides of a polygon taken in order, then their resultant is zero and hence the forces will be in equilibrium.

Let the forces acting on a body at a point O be as below:

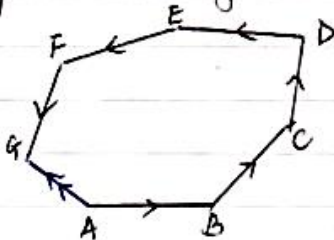


The vector sum of these forces is zero and the system is in equilibrium.

This too holds for any number of forces. Also the forces need not be in one plane.

Resultant of any number of forces.

Consider a number of forces acting on a body at a point O. We draw a polygon with its sides proportional and parallel to these forces. If the polygon closes, we say that the system of forces is in equilibrium, but if it does not close, then the resultant is represented by the straight line needed to close the figure.

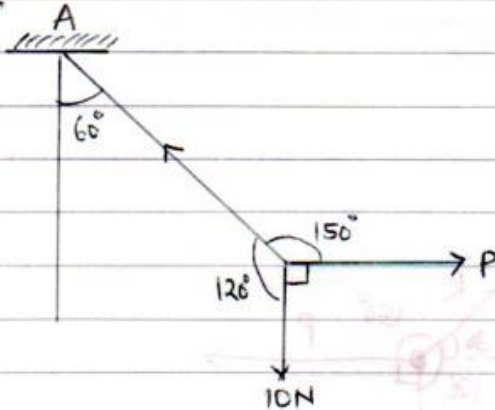


$$\vec{AG} = \vec{AB} + \vec{BC} + \vec{CD} + \vec{DE} + \vec{EF} + \vec{FG}$$

Example 6

One end of a string is fixed to a point A and an object of weight 10 N is fastened to the other end. The object is pulled aside by a horizontal force until the string is inclined at 60° to the vertical at A. Calculate the magnitudes of the tension in the string and the horizontal force.

Sol:



Using Lami's theorem,

$$\frac{10}{\sin 150} = \frac{P}{\sin 120} = \frac{T}{\sin 90}$$

$$T = \frac{10}{\sin 150} \times \sin 90 = \underline{\underline{20\text{ N}}}$$

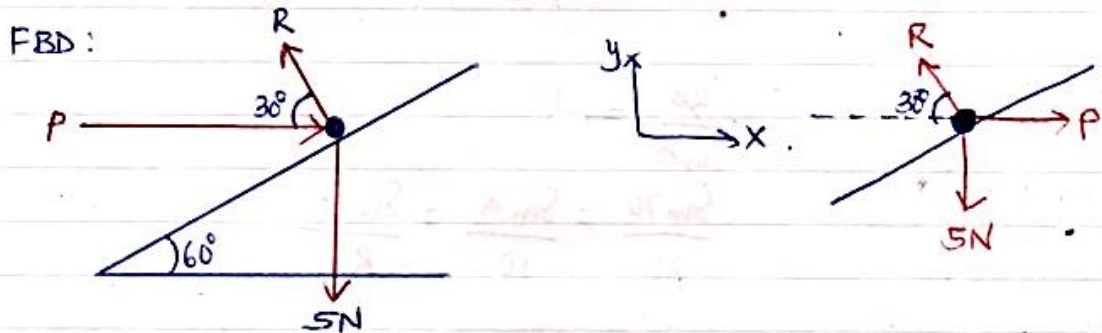
$$\therefore P = \underline{\underline{10\sqrt{3}\text{ N}}}$$

Example 7:

A 500g particle stands on a smooth plane inclined at 60° to the horizontal. Find the least horizontal force which prevents it from sliding and the corresponding reaction between the particle and the plane.

Sol:

Smooth plane $\Rightarrow$ no friction



Resolving in x & y

$$R: \begin{pmatrix} -R \cos 30^\circ \\ R \sin 30^\circ \end{pmatrix} \quad 5N: \begin{pmatrix} 0 \\ -5N \end{pmatrix} \quad P: \begin{pmatrix} +P \\ 0 \end{pmatrix}$$

Equating x -components (horizontal)

$$-R \cos 30^\circ + P = 0$$

$$P = R \cos 30^\circ$$

Equating y -components (vertical)

$$R \sin 30^\circ - 5 = 0 \Rightarrow R \sin 30^\circ = 5$$

$$\therefore R = \underline{10N}$$

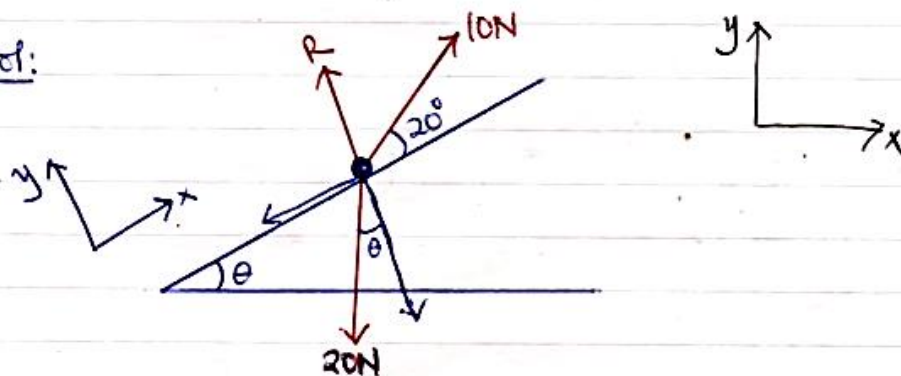
$$\therefore P = R \cos 30^\circ = 10 \cos 30^\circ = 5\sqrt{3} N$$

$$\underline{P = 8.660 N.}$$

Example 8:

A 2kg particle lies on a smooth plane. Given that a force of 10N acting away from the plane at 20° to the line of greatest slope, is just sufficient to prevent it from sliding, find the angle of inclination of the plane and the reaction between it and the particle.

Sol:



Get components in x and y as indicated

$$10N: \begin{pmatrix} 10 \cos 20^\circ \\ 10 \sin 20^\circ \end{pmatrix} \quad 20N: \begin{pmatrix} -20 \sin \theta \\ -20 \cos \theta \end{pmatrix} \quad R: \begin{pmatrix} 0 \\ R \end{pmatrix}$$

Equating components:

$$X: 10 \cos 20 - 20 \sin \theta = 0$$

$$\text{i.e. } \sin \theta = \frac{10 \cos 20}{20}$$

$$\Rightarrow \theta = \underline{\underline{28.02^\circ}}$$

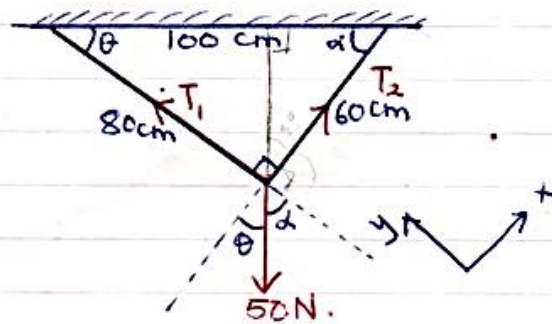
$$Y: 10 \sin 20 - 20 \cos \theta + R = 0$$

$$\begin{aligned} \therefore R &= 20 \cos 28.02 - 10 \sin 20 \\ &= \underline{\underline{14.23N}} \end{aligned}$$

Example 9:

A 5kg particle is attached by two strings of lengths 60 cm and 80 cm to two points on a horizontal ceiling which are 1m apart. Find the tensions in the strings.

Sol
FBD:



~~8m~~ 8m

$$Y: T_1 - 50 \cos \alpha = 0$$

$$T_1 = 50 \cos \alpha \quad \text{but } \cos \alpha = 6/10 = 3/5$$

$$= 50 \times 3/5$$

$$= \underline{\underline{30 \text{ N}}}$$

$$X: T_2 - 50 \sin \alpha = 0$$

$$T_2 = 50 \sin \alpha \quad \text{but } \sin \alpha = 8/10 = 4/5$$

$$= 50 \times 4/5$$

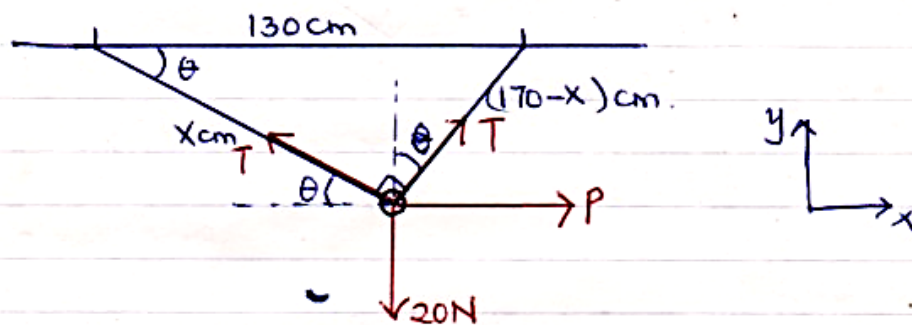
$$= \underline{\underline{40 \text{ N}}}$$

Ex: Try using Lami's theorem.

Example 10:

A string of length 170 cm is attached at its ends to two points at the same level which are 130 cm apart. The string carries a smooth ring of mass 2 kg. A horizontal force is applied to the ring, causing the angle between the two sections of string to be 90° . Find this force and the tension in the string.

Sol:



Using Pythagoras Thm:

$$x^2 + (170-x)^2 = 130^2$$

$$x^2 + 28,900 - 340x + x^2 = 16,900$$

$$2x^2 - 340x + 12,000 = 0$$

$$x^2 - 170x + 6000 = 0$$

$$(x-50)(x-120) = 0$$

$$x = 120, \text{ or } 50.$$

Acc. to diagram $x = 120$ $\therefore$ other side is 50.

i.e. we have a 5, 12, 13 right-angled triangle.

$$\text{Thus } \sin \theta = 5/13 \quad \cos \theta = 12/13.$$

$$\text{Resolving in } y: T \sin \theta + T \cos \theta - 20 = 0 \quad \text{--- (i)}$$

$$x: -T \cos \theta + T \sin \theta + P = 0 \quad \text{--- (ii)}$$

$$\text{Solving (i): } T(5/13 + 12/13) = 20$$

$$\Rightarrow T = \underline{\underline{15.29 \text{ N}}}$$

Solving (ii) $P = T \cos \theta - T \sin \theta$
 $= 15.29 (12/13 - 5/13)$
 $= \underline{\underline{8.233 \text{ N}}}$

Ex:

- 1) A string ABC has a body of mass 10kg attached to B, and the ends A and C are attached to two points on a horizontal ceiling. If angle BAC and ACB are 50° and 60° respec., what are the tensions in AB and BC?
 (Ans: $T_1 = 53.21 \text{ N}$, $T_2 = 68.40 \text{ N}$.)
- 2) The forces $(3\hat{i} + 5\hat{j})\text{N}$, $(a\hat{i} + b\hat{j})\text{N}$, $(8\hat{i} - 6\hat{j})\text{N}$ and $(-4\hat{i} - 3\hat{j})\text{N}$ are in equilibrium. Find the values of a and b.
 ($a = -7$, $b = 4$).